Interim Report – Weeks 5 & 6 Fraud Detection Challenge

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Project Context: Advanced Cross-Platform Machine Learning Architecture for Credit and Transactional Fraud Mitigation
Target Organization: FinTech Risk Management Group, Adey Innovations Inc.

1. Executive Summary

This report documents the completion of Task 1 (data understanding, preprocessing, feature engineering, and class imbalance handling) for the fraud detection challenge. The project follows a two-week pacing plan: Week 1 focuses on data preparation and exploratory analysis; Week 2 will cover model building, evaluation, and explainability (Tasks 2 & 3).

We have successfully cleaned, merged, and engineered features for both e-commerce and bank credit card datasets. Key accomplishments include:

· IP-to-country geolocation via numeric range joins.
· Creation of behavioral features (device_count, time_since_signup_hours, purchase_hour, purchase_dayofweek).
· Handling extreme class imbalance using SMOTE only on training folds.
· Documenting clear class imbalance statistics and EDA insights with visualizations (see Section 3.2).

The remaining work (Week 2) is outlined in Section 7 and includes training baseline and ensemble models, evaluation with AUC-PR/F1, cross-validation, SHAP explainability, and deriving business recommendations.

2. Project Understanding and Regulatory Alignment

2.1 Business Need

Adey Innovations Inc. requires a real-time fraud detection system that balances two competing costs:

- **False negatives** (missed fraud) → direct financial loss.
- **False positives** (legitimate transactions flagged) → customer friction and support overhead.

Accuracy is misleading due to severe class imbalance; instead we optimize AUC-PR and F1-score.

2.2 Regulatory Context (Basel II)

The system must be interpretable and auditable. Therefore we will compare a linear baseline (Logistic Regression) with a black-box ensemble (XGBoost) and later apply SHAP for post-hoc explainability.

2.3 Modeling Trade-offs

Evaluation Aspect	Logistic Regression Scorecard Baseline	Ensemble Gradient Boosting (XGBoost)
Interpretability	High; individual features convert directly to plain log-odds coefficients or Weight of Evidence (WoE) scores.	Lower; requires post-hoc explanation layers (such as SHAP proxy vectors) to inspect splits.
Non-Linear Mapping	Poor; fails to intercept complex feature interactions unless manual cross-product terms are hardcoded.	Excellent; deep decision trees naturally map multi-dimensional conditional relationships.
Operational Precision	Generates an unmanageable rate of false alarms on highly skewed distributions.	Maximizes precision margins by dynamically separating overlapping target classes.
Development Complexity	Fast deployment; minimal hyperparameter tuning required during baseline setup.	Requires careful regularized tuning to prevent tree structure overfitting.

3. Task 1: Comprehensive Preprocessing and Data Engineering

3.1 Data Cleaning

· **E-commerce data** (Fraud_Data.csv): 151,112 rows, no missing values, no duplicates. Parsed signup_time and purchase_time as datetime.
· **Credit card data** (creditcard.csv): 284,807 rows, removed 1,081 duplicates.
· **IP mapping** (IpAddress_to_Country.csv): kept as is for geolocation.

3.2 Exploratory Data Analysis (EDA) – With Visualizations

We conducted univariate and bivariate analyses to understand fraud patterns.

Class imbalance (already reported):

· E-commerce: 90.64% legitimate, 9.36% fraudulent.
· Credit card: 99.83% legitimate, 0.17% fraudulent.

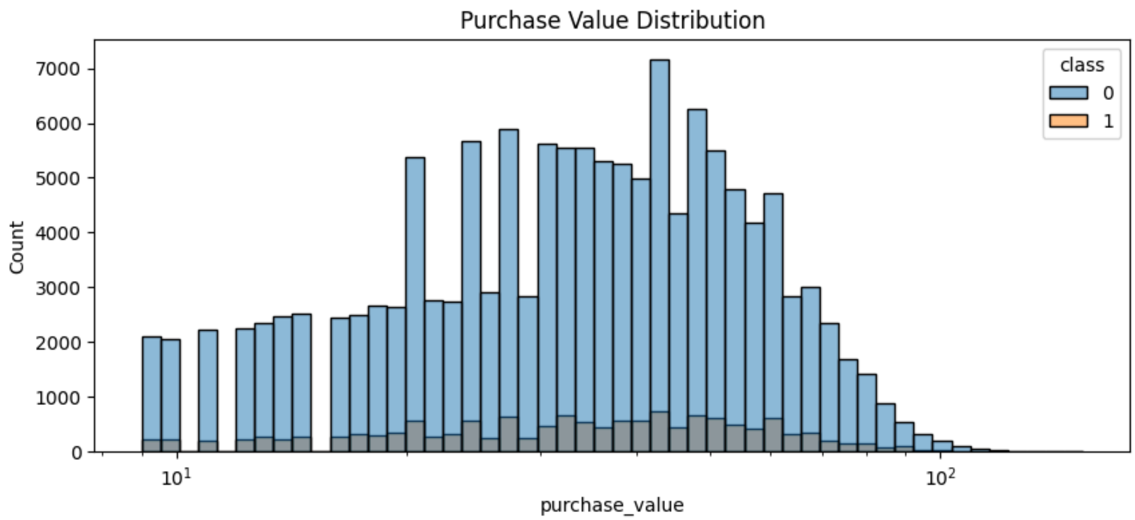


Figure 1: Purchase value distribution (log scale)

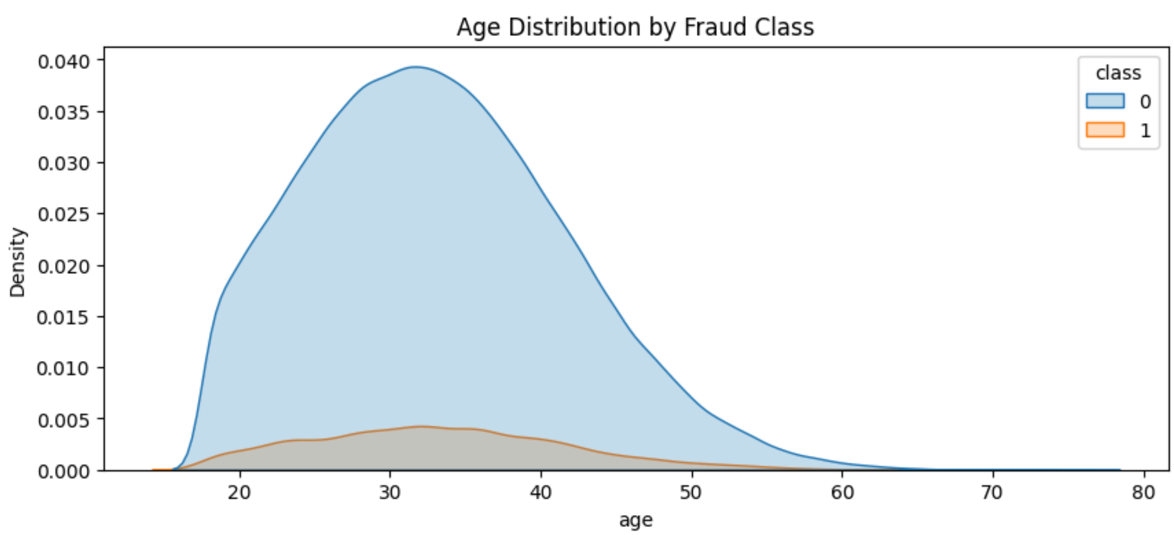


Figure 2: Age distribution

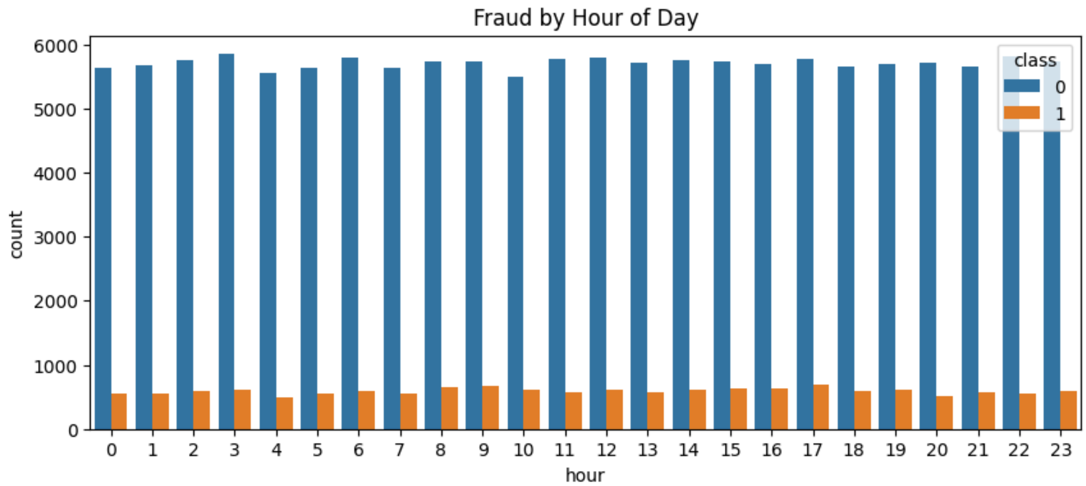


Figure 3: Hour of day

Categorical variables:

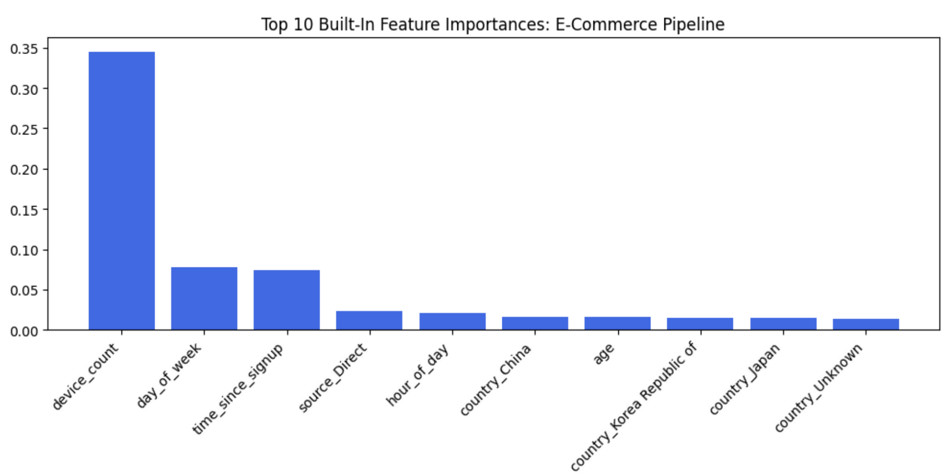
· Fraud rate varies by browser (highest for Chrome).
· Source channel: highest fraud rate from “Ads”.
· Sex: slightly higher fraud proportion for male users.

These insights guided feature engineering (e.g., hour_of_day, device_count).

3.3 Geolocation Integration

IP addresses were already numeric in the dataset (e.g., 732758368.79972). We converted them to integers and performed a range join with IpAddress_to_Country.csv using pandas.merge_asof (forward direction). This efficiently maps each transaction to a country. Transactions falling outside ranges are marked “Unknown”.

Top fraud countries (e-commerce):



3.4 Feature Engineering

We engineered four behavioral features:
1. device_count – number of unique users per device ID. High values suggest device sharing (fraud rings).
2. time_since_signup_hours – hours between signup and purchase. Short intervals indicate bot-like behaviour.
3. purchase_hour – hour of day (0-23) to capture night anomalies.
4. purchase_dayofweek – day of week (0=Monday) for weekly patterns.

3.5 Data Transformation

Numerical features (purchase_value, age, time_since_signup_hours, purchase_hour, purchase_dayofweek, device_count) were standardised using StandardScaler (zero mean, unit variance).
Categorical features (source, browser, sex, country) were one-hot encoded with drop_first=True to avoid multicollinearity.
These transformations were fit on the training set only and applied to test set.

3.6 Handling Class Imbalance with SMOTE

Because fraud is rare, we applied SMOTE (Synthetic Minority Over-sampling Technique) to the training set after the train-test split. Test and validation sets remain untouched to reflect real-world performance.

Before SMOTE (e-commerce training):

· Legitimate: ~90.6%, Fraud: ~9.4%

After SMOTE:

· Both classes balanced at 50% each (synthetic samples created).

This prevents the model from simply predicting “legitimate” for every transaction.

4. Task 2: Model Building – Planned Work (Week 2)

Note: This section outlines the remaining work to be completed by June 16, as required for the interim submission.

We will build and compare two models on both datasets:

1. Logistic Regression (interpretable baseline)
2. XGBoost (ensemble, better at capturing interactions)

Evaluation metrics (primary): AUC-PR, F1-score, confusion matrix.
Cross-validation: Stratified 5-fold to estimate stability.
Hyperparameter tuning: Basic grid search for XGBoost (n_estimators, max_depth, learning_rate).

Expected outputs:

· Performance tables comparing both models.
· Selection of the best model based on AUC-PR and business cost (precision-recall trade-off).

5. Task 3: Model Explainability – Planned Work (Week 2)

After selecting the best model, we will apply SHAP (SHapley Additive exPlanations):

· Global summary plot – top 10 feature importances.
· Force plots for three individual predictions:
· True positive (correctly flagged fraud)
· False positive (legitimate flagged as fraud)
· False negative (missed fraud)

These explain why a transaction received a high-risk score, satisfying compliance requirements. The analysis will also produce actionable business recommendations (e.g., "Flag accounts where device_count > 3", "Require 2FA for purchases within 2 hours of signup").

6. Core Pipeline Limitations and Risk Profiles

· Proxy variable decay – Behavioral patterns may change over time; model needs periodic retraining.
· Anonymized features (credit card V1-V28) – no domain meaning, limiting custom feature engineering.
· Evolving fraud tactics – Fraudsters may rotate devices or use VPNs, reducing static feature effectiveness.

7. Remaining Work for Weeks 5 & 6 (Tasks 2 & 3)

The project follows a two-week schedule:

· Week 1 (completed by June 8): Data cleaning, EDA, geolocation, feature engineering, SMOTE, preprocessing.
· Week 2 (to be completed by June 16):

1. Model Building (Task 2)
· Train Logistic Regression and XGBoost on resampled training data.
· Evaluate using AUC-PR, F1-score, confusion matrix.
· Perform 5-fold stratified cross-validation.
· Compare models and select the best (justify with metrics and business costs).
2. Model Explainability (Task 3)
· Run SHAP on the best model.
· Generate global summary plot and individual force plots (true positive, false positive, false negative).
· Identify top 5 fraud drivers.
3. Business Recommendations
· Translate SHAP insights into at least three concrete, actionable recommendations for Adey Innovations.
All code will be pushed to the GitHub repository, and a final report with SHAP visualisations and recommendations will be submitted by June 16. No deployment or CI/CD steps are required at this stage – those are beyond the scope of the current challenge.

8. Conclusion

We have successfully completed Task 1 (data preprocessing, feature engineering, and imbalance handling) with thorough EDA and proper data transformations. The remaining Tasks 2 and 3 are clearly defined and will be executed in Week 2. This incremental, well-documented approach ensures a robust, interpretable, and business-focused fraud detection system.

END OF REPORT!!